

Exercise4

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Exercise 4

Use simulation to solve this exercise.

From a pack of 52 cards a card is drawn and not replaced. A second card is drawn. What is the probability that the first card is an ace and the second card is a king?

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In [13]:
from numpy.random import default_rng
from itertools import product
deck = list(product(['He', 'Cl', 'Di', 'Sp'], [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]))
rng = default_rng()
n_shuffles = 100000
count = 0
for i in range(n_shuffles):
    cards_drawn = rng.choice(deck, size = 2, replace = False)
    if (
        (all(cards_drawn[0] == ('Di', 1)) or
         all(cards_drawn[0] == ('He', 1)) or
         all(cards_drawn[0] == ('Cl', 1)) or
         all(cards_drawn[0] == ('Sp', 1)))
        and
        (all(cards_drawn[1] == ('Di', 13)) or
         all(cards_drawn[1] == ('He', 13)) or
         all(cards_drawn[1] == ('Cl', 13)) or
         all(cards_drawn[1] == ('Sp', 13)))
    ):
        count += 1

print(f"The probability that the first card is ace and the second is king is {count / n_shuffles:.5f}")
print(f"Theoretical probability = {(4/52)*(4/51):.5f}")
```

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Out [13]:
The probability that the first card is ace and the second is king is 0.00599
Theoretical probability = 0.00603
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